Variational quantum eigensolver

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Guess,
measure,
improve

Variational Principle

For any trial state ψ, the energy ⟨ψ|H|ψ⟩ is at least E_0.

So the lower the energy we find, the better the guess.

Minimizing over a family of states gives an upper bound on E_0.

The quality depends on the family we choose.

$$E_{\text{approx}} = \frac{\langle \Psi_{\text{trial}} | \hat{H} | \Psi_{\text{trial}} \rangle}{\langle \Psi_{\text{trial}} | \Psi_{\text{trial}} \rangle} \geq E_0$$

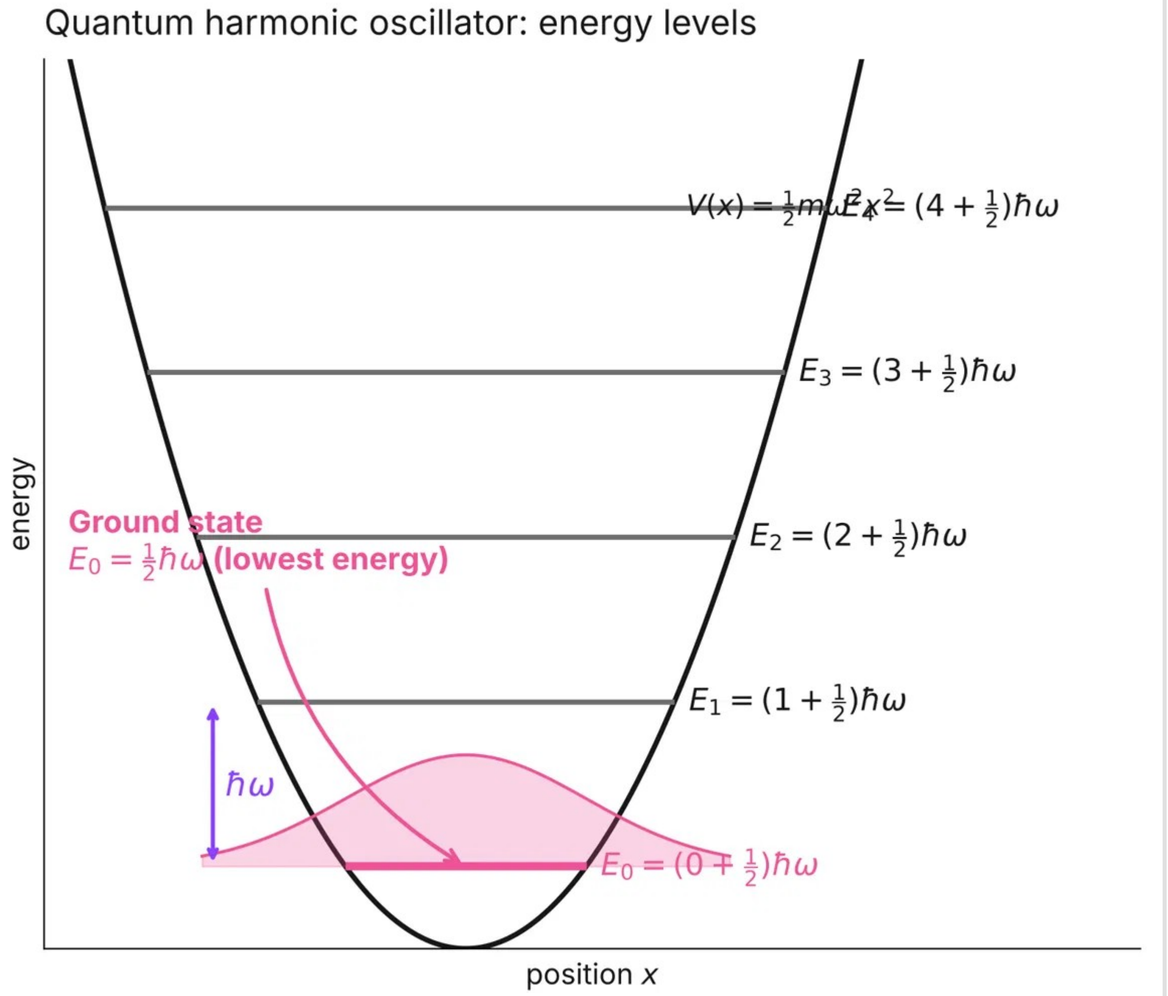
The problem: find the ground state

The ground state is the lowest-energy state of a system.

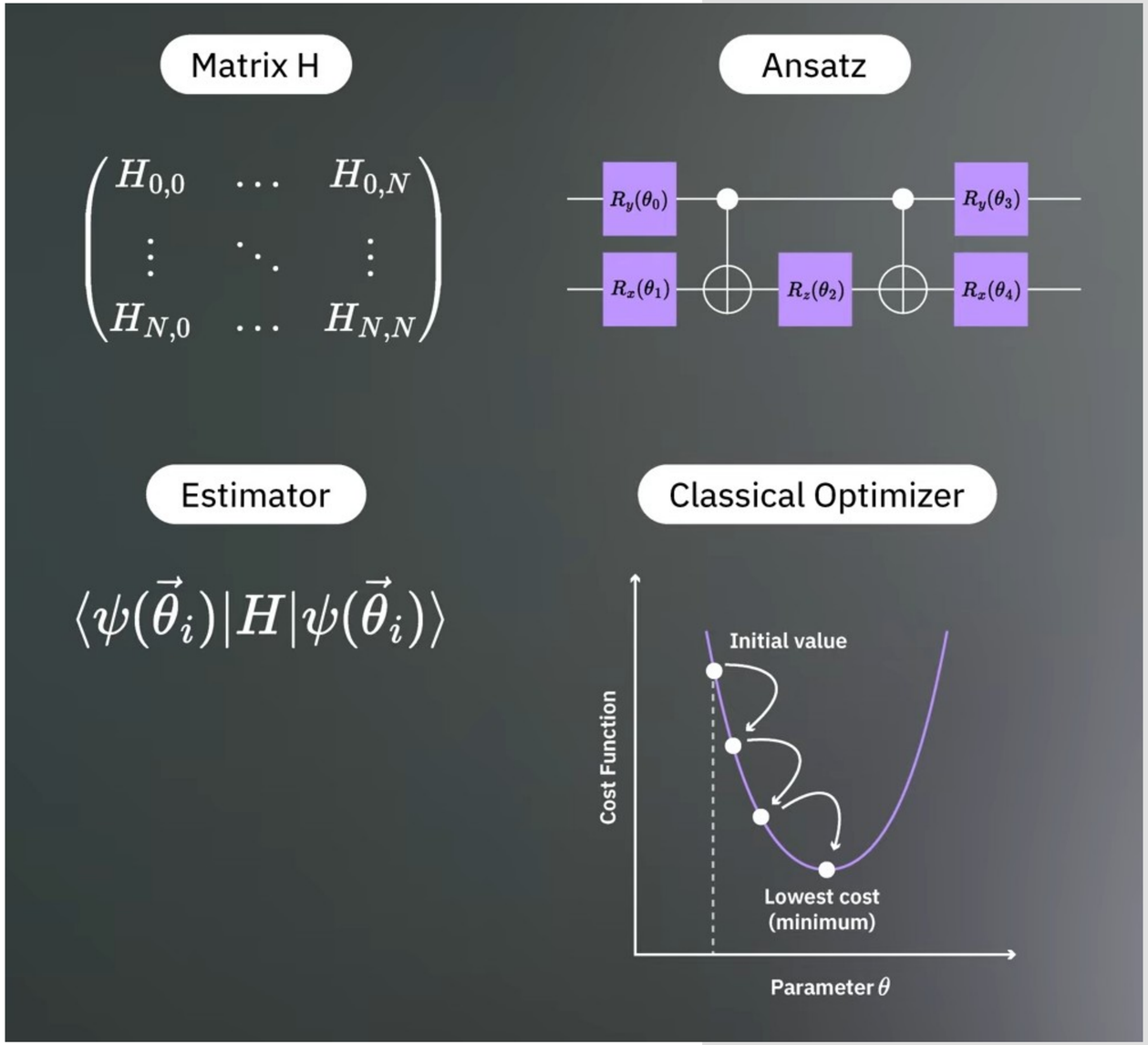
It decides the structure and bonds of a molecule.

Goal: the ground state energy E_0 of a Hamiltonian H.

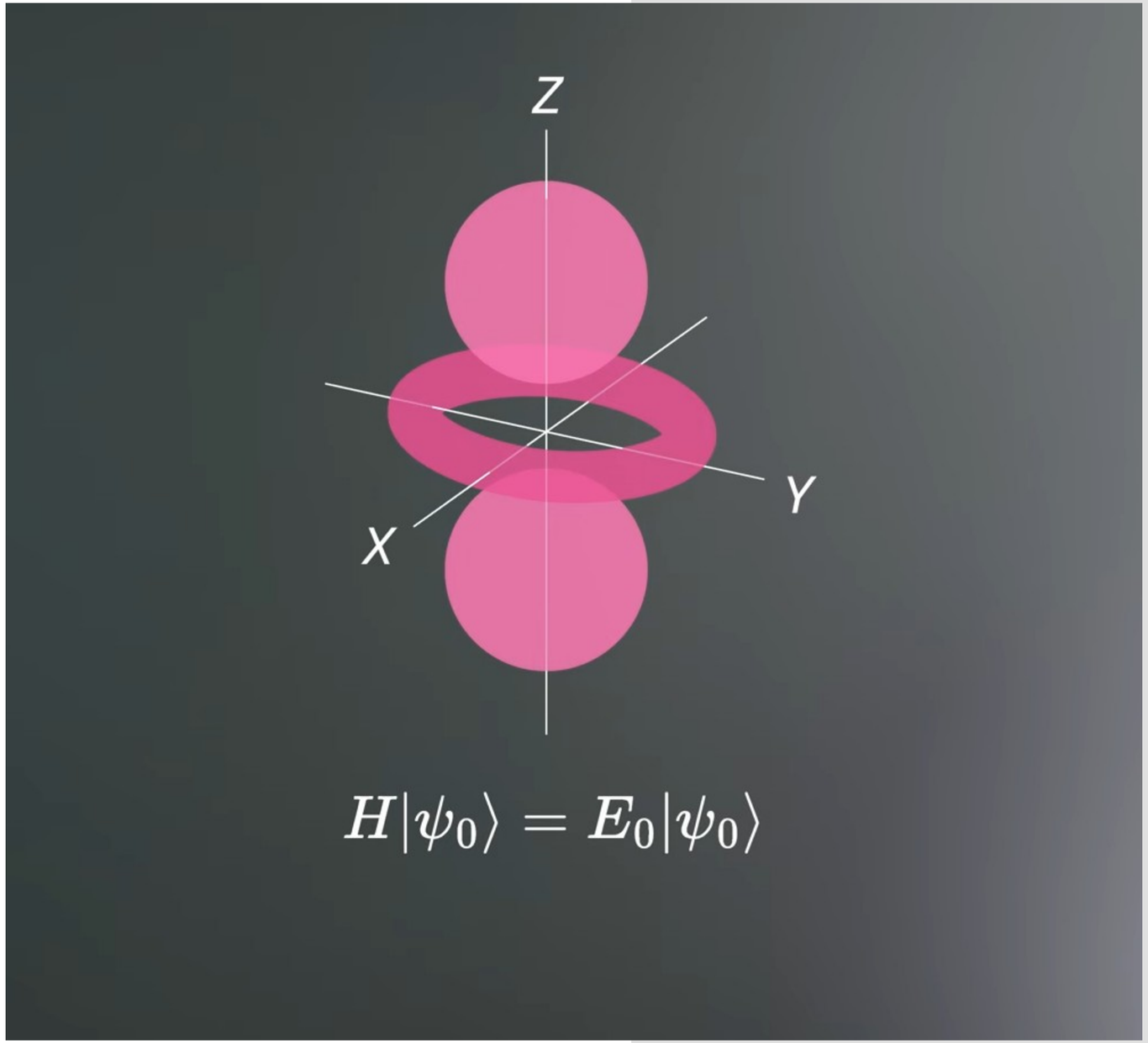
We study H atoms and the H_2 molecule.



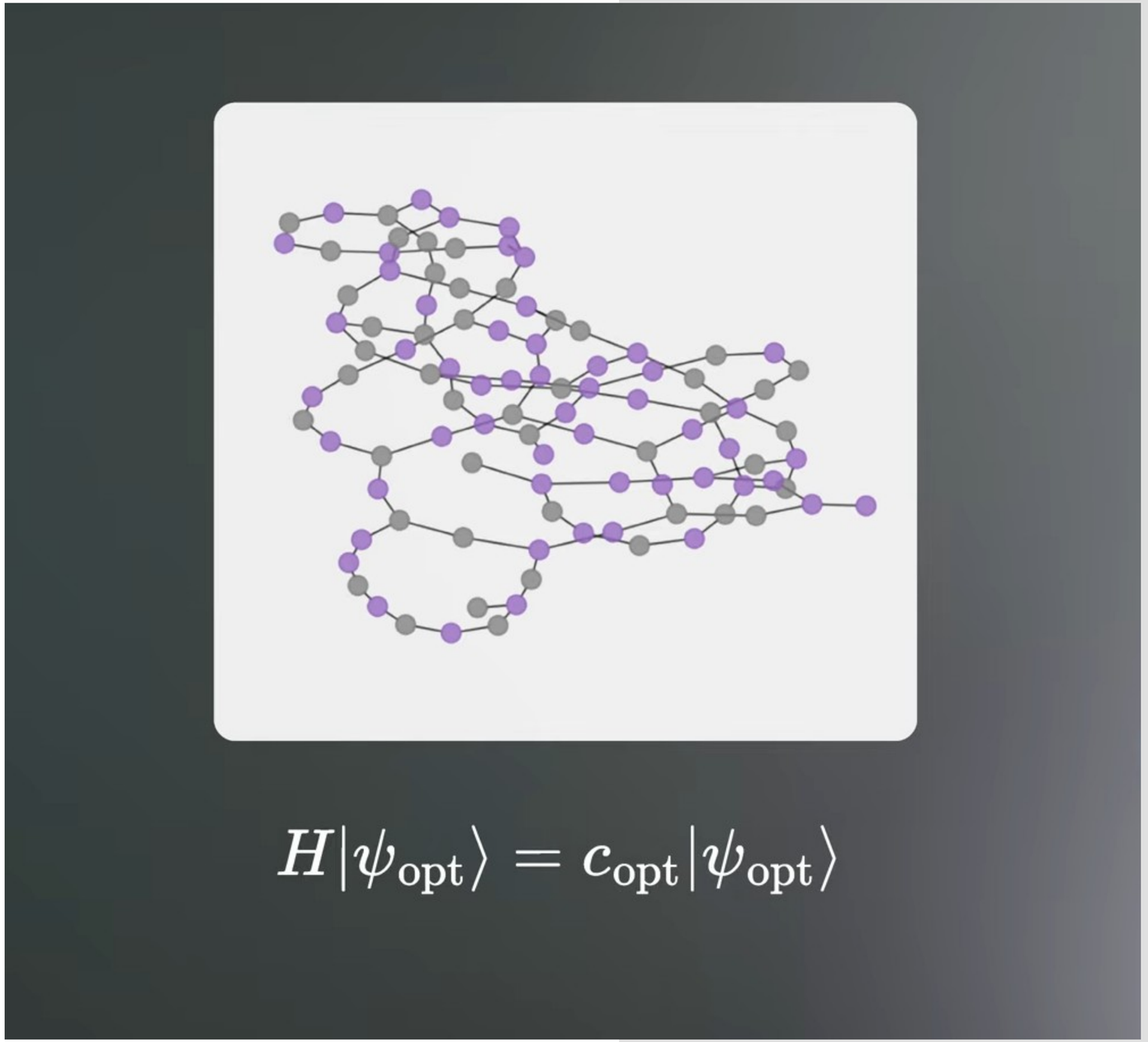
The Work Flow



The Operator

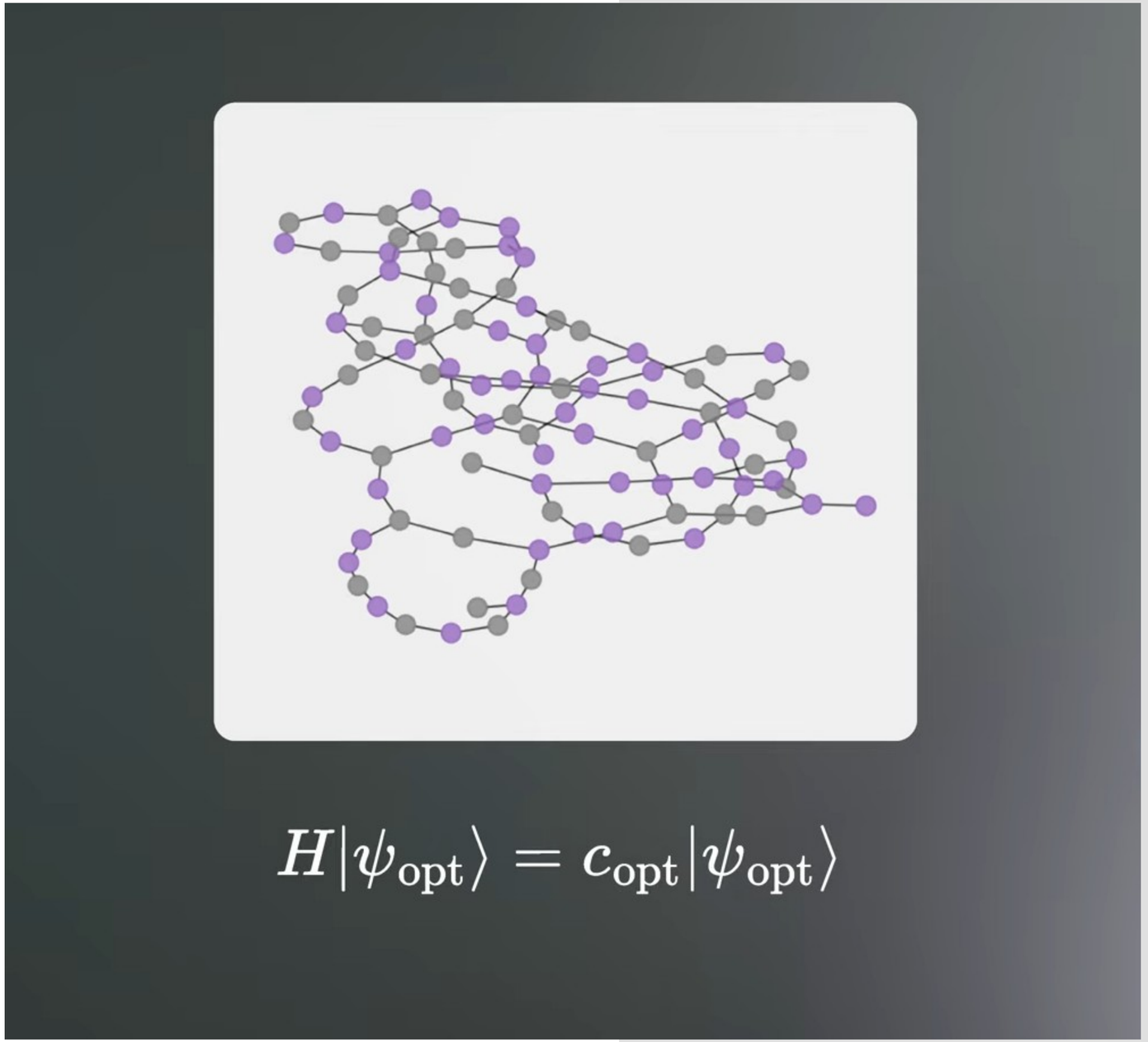


The Operator



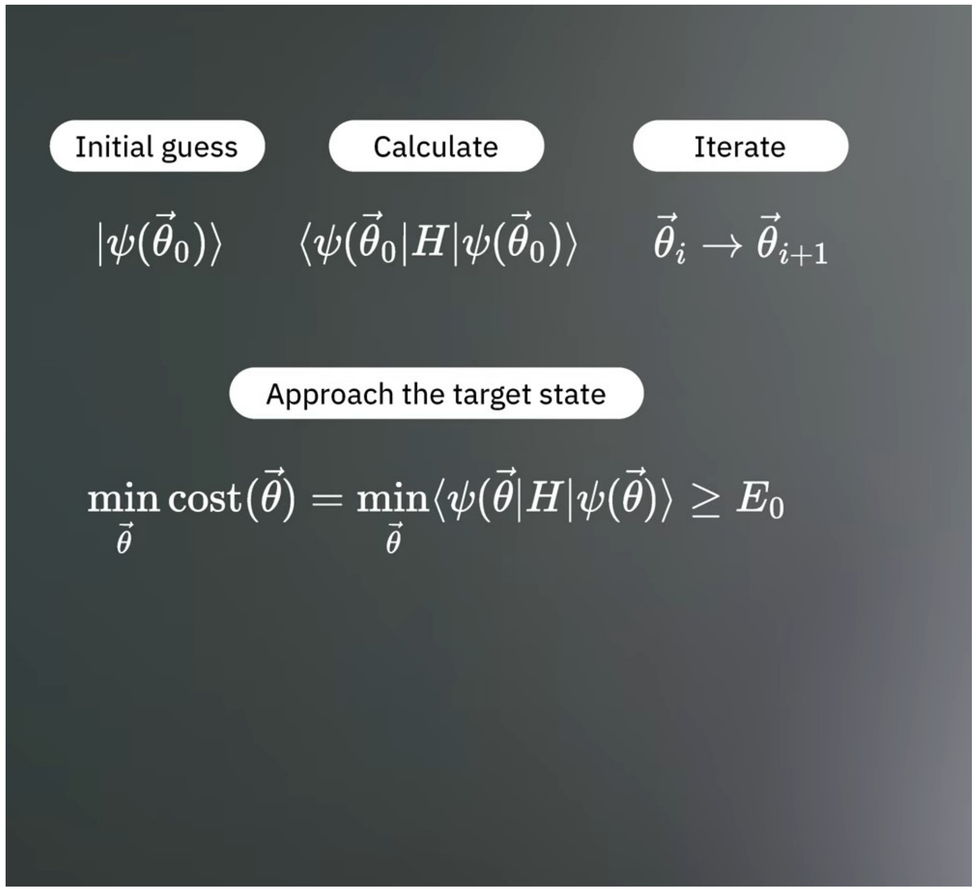
$$H|\psi_{\text{opt}}\rangle = c_{\text{opt}}|\psi_{\text{opt}}\rangle$$

The Operator



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Two computers, one loop



Quantum computer

Prepares a trial state with a parameterized circuit.

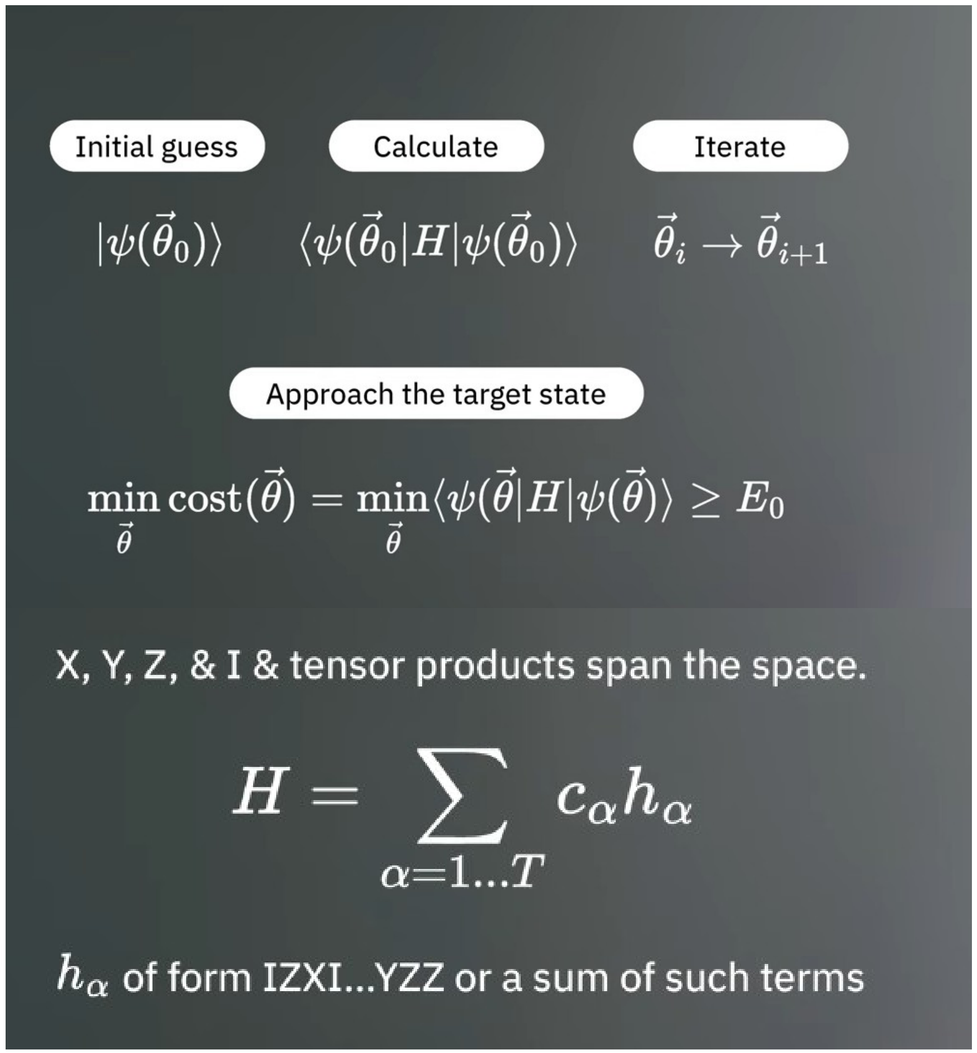
Measures the energy of that state.

Classical computer

Reads the energy.

Chooses better parameters and repeats.

Two computers, one loop



Quantum computer

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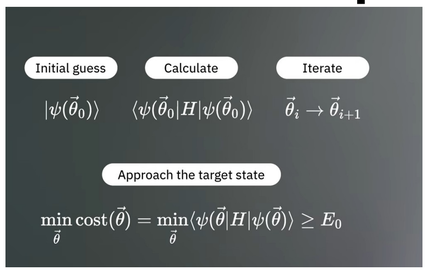
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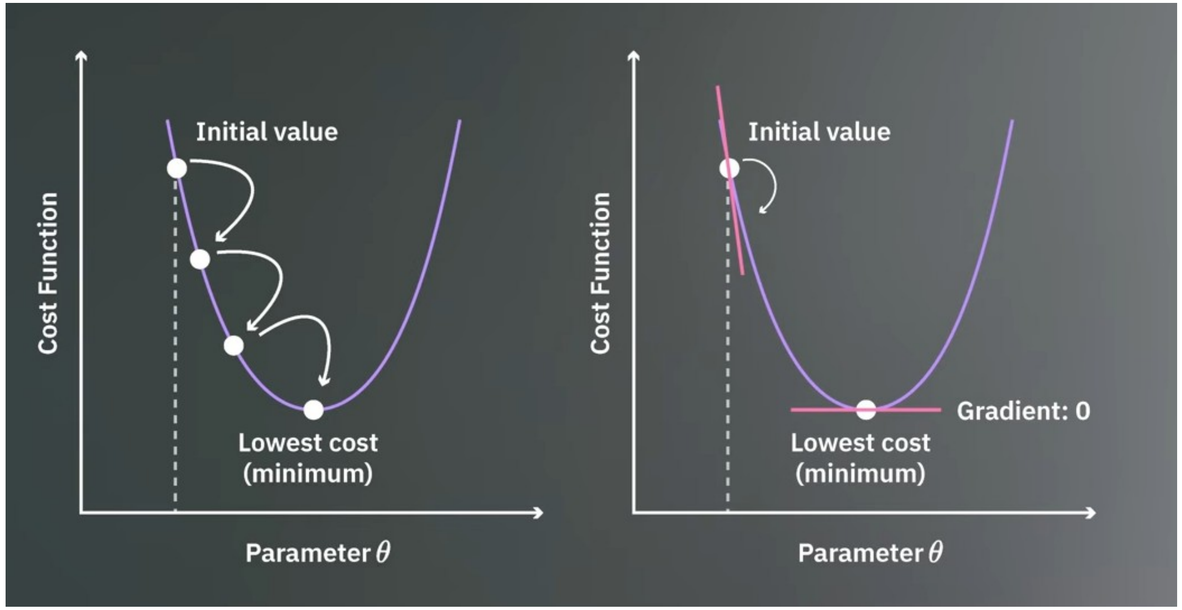
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Quantum computer

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Measures the energy of that state.

Classical computer

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What we learned

1

Variational principle. Any trial state gives an upper bound on the ground-state energy.

2

Hybrid loop. The quantum computer measures and the classical computer optimizes.

3

Ansatz and optimizer matter. More parameters mean more reach, but harder optimization.

Adapted from IBM Quantum Learning: Variational quantum eigensolver

Will it magically solve your problems

Classical variation

- → All components classical
- → Cost scales like $O(N^2)$ worst case
- → Requires good initial guess

VQE

- → Only the optimizer is purely classical
- → Cost scales with precision, N_{GCP}
- → Requires good initial guess

Now, open the VQE notebook

